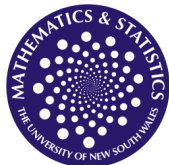




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Lecture Week 2: Priors and Inversion Sampling

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Outline

1. Different types of priors
 - Conjugate priors, improper priors, Jeffreys' priors
2. Monte Carlo Methods
 - Inversion sampling.

Outline

1. Different types of priors
 - Conjugate priors, improper priors, Jeffreys' priors
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Summary: Bayesian updating

There are 4 key steps in the Bayesian approach:

- ▶ Specification of a likelihood model $L(x|\theta)$;
- ▶ Determination/elicitation of a suitable prior distribution $\pi(\theta)$;
- ▶ Calculation of the posterior distribution via Bayes' Theorem

$$\pi(\theta|x) = \frac{L(x|\theta)\pi(\theta)}{\int_{\theta} L(x|\theta)\pi(\theta)d\theta}$$

- ▶ Draw inference from the posterior.

Effect of prior on posterior computation

In order to compute probabilities based on the posterior

$$\pi(\theta|x) = \frac{L(x|\theta)\pi(\theta)}{\int_{\theta} L(x|\theta)\pi(\theta)d\theta} = \frac{L(x|\theta)\pi(\theta)}{m(x)}$$

we need to be able to evaluate the normalising constant, $m(x)$.

Example:

Suppose $x_1, \dots, x_n \sim \text{Poisson}(\theta)$ and that we believe $0 < \theta \leq 1$ with all values equally likely. Hence $\pi(\theta) = 1, 0 < \theta \leq 1$.

Here the normalising constant is:

$$m(x) = \int_0^1 \exp(-n\theta) \theta^{\sum x_i} d\theta$$

which can only be evaluated numerically.

The point:

Even simple priors can lead to awkward numerical problems.
Issue is magnified as $\dim(\theta)$ increases.

Conjugate priors

Conjugate prior definition:

If

- ▶ $\mathcal{F}$ is a class of sampling distributions (likelihoods) $L(x|\theta)$, and
- ▶ $\mathcal{P}$ is a class of prior distributions $\pi(\theta)$ for θ ,

then the class $\mathcal{P}$ is **conjugate** for $\mathcal{F}$ if

$$\pi(\theta|x) \in \mathcal{P} \quad \text{for all } L(\cdot|\theta) \in \mathcal{F} \text{ and } \pi(\cdot) \in \mathcal{P}.$$

i.e. Conjugate priors permit the posterior distribution to **stay in the same family as the prior**.

Viable if family $\mathcal{P}$ is flexible enough to describe prior beliefs.

Conjugate priors

Example #1: Binomial model

Suppose we have $x \sim \text{Binomial}(n, \theta)$ and we wish to make inference on $0 \leq \theta \leq 1$.

Here

$$L(x|\theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x = 0, \dots, n.$$

While $\pi(\theta)$ will vary from problem to problem, we want to choose a prior family that will **simplify posterior computations**.

Specify $\theta \sim \text{Beta}(a, b)$ so that

$$\begin{aligned} \pi(\theta) &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} & 0 \leq \theta \leq 1 \\ &\propto \theta^{a-1} (1-\theta)^{b-1}. \end{aligned}$$

Conjugate priors

Example #1: Binomial model (cont.)

Then, by Bayes' Theorem:

$$\begin{aligned}\pi(\theta|x) &\propto L(x|\theta)\pi(\theta) \\ &\propto \theta^x(1-\theta)^{n-x} \times \theta^{a-1}(1-\theta)^{b-1} \\ &= \theta^{a+x-1}(1-\theta)^{b+n-x-1}.\end{aligned}$$

Since we know $\pi(\theta|x)$ must be a proper density function ($\int \pi(\theta|x)d\theta = 1$), then we must have

$$\theta|x \sim \text{Beta}(a+x, b+n-x).$$

- ▶ Effect of data: modify Beta parameters from (a, b) to $(a+x, b+n-x)$;
- ▶ Staying within Beta family means posterior computation is easy.

Conjugate priors

Example #2: Poisson model

Suppose $x_1, \dots, x_n$ are *iid* $\text{Poisson}(\theta)$ variables.

Then

$$\begin{aligned} L(x|\theta) &= \prod_{i=1}^n \exp(-\theta) \theta^{x_i} / x_i! \\ &= \exp(-n\theta) \theta^{\sum x_i} / \prod_{i=1}^n x_i!. \end{aligned}$$

Suppose prior beliefs can be represented by $\theta \sim \text{Gamma}(a, b)$, so that

$$\begin{aligned} \pi(\theta) &= \frac{a^b}{\Gamma(a)} \theta^{a-1} \exp(-b\theta) \quad \theta > 0 \\ &\propto \theta^{a-1} \exp(-b\theta). \end{aligned}$$

Conjugate priors

Example #2: Poisson model (cont.)

Then by Bayes' Theorem:

$$\begin{aligned}\pi(\theta|x) &\propto L(x|\theta)\pi(\theta) \\ &\propto \exp(-n\theta)\theta^{\sum x_i}\theta^{a-1}\exp(-b\theta) \\ &= \theta^{a+\sum x_i-1}\exp(-(b+n)\theta).\end{aligned}$$

Hence, we must have

$$\theta|x \sim \text{Gamma}(a + \sum x_i, b + n).$$

- ▶ Effect of data: modify Gamma parameters from (a, b) to $(a + \sum x_i, b + n)$;
- ▶ Stays within Gamma family.

Conjugate priors

Example #3: Gamma model

Suppose $x_1, \dots, x_n$ are *iid* $\text{Gamma}(k, \theta)$ variables, with k known/fixed. Then

$$\begin{aligned} L(x|\theta) &= \prod_{i=1}^n \frac{\theta^k}{\Gamma(k)} x_i^{k-1} \exp(-x_i \theta) \\ &= \theta^{kn} \exp(-\theta \sum x_i) \times \frac{\prod_{i=1}^n x_i^{k-1}}{\Gamma(k)^n} \end{aligned}$$

Suppose prior beliefs can again be represented by $\theta \sim \text{Gamma}(a, b)$:

$$\pi(\theta) \propto \theta^{a-1} \exp(-b\theta).$$

Then by Bayes' Theorem:

$$\begin{aligned} \pi(\theta|x) &\propto \theta^{kn} \exp(-\theta \sum x_i) \theta^{a-1} \exp(-b\theta) \\ &= \theta^{a+kn-1} \exp(-(b + \sum x_i)\theta) \\ &\sim \text{Gamma}(a + kn, b + \sum x_i). \end{aligned}$$

Conjugate priors

When can conjugate priors be found?

A: The only easy case is for models in the exponential family:

$$f(x|\theta) = h(x)g(\theta) \exp\{t(x)c(\theta)\}$$

for functions h, g, t and c such that $\int f(x|\theta)dx = 1$.

Includes: Exponential, Poisson, one-parameter Gamma, Binomial, Normal (with known variance).

There are a few examples outside the exponential family (e.g. $U(0, \theta)$ likelihood and Pareto prior on θ) (out of scope of this course)

Conjugate priors

In this case, with prior $\pi(\theta)$, we have:

$$\begin{aligned}\pi(\theta|x) &\propto L(x|\theta)\pi(\theta) \\ &= \pi(\theta) \prod_{i=1}^n h(x_i)g(\theta) \exp\{t(x_i)c(\theta)\} \\ &\propto \pi(\theta)g(\theta)^n \exp\{c(\theta) \sum t(x_i)\}.\end{aligned}$$

Which means that if we choose

$$\pi(\theta) \propto g(\theta)^d \exp\{bc(\theta)\},$$

for some d and b , then we obtain

$$\begin{aligned}\pi(\theta|x) &\propto g(\theta)^{d+n} \exp\{c(\theta)[\sum t(x_i) + b]\} \\ &= g(\theta)^{\tilde{d}} \exp\{\tilde{b}c(\theta)\},\end{aligned}$$

which is in the exponential family (with modified parameters).

Conjugate priors

Example : Binomial model

Here we have

$$\begin{aligned}f(x|\theta) = L(x|\theta) &= \binom{n}{x} \theta^x (1-\theta)^{n-x}, \\&= \binom{n}{x} (1-\theta)^n \exp \left\{ x \log \left(\frac{\theta}{1-\theta} \right) \right\}.\end{aligned}$$

So here we have: $h(x) = \binom{n}{x}$, $g(\theta) = (1-\theta)^n$, $t(x) = x$ and $c(\theta) = \log \left(\frac{\theta}{1-\theta} \right)$.

Hence we need to construct a prior of the form

$$\begin{aligned}\pi(\theta) &\propto [(1-\theta)^n]^d \exp \left\{ b \log \left(\frac{\theta}{1-\theta} \right) \right\} \\&= (1-\theta)^{nd-b} \theta^b,\end{aligned}$$

which is in the Beta family.

Conjugate priors

Example: Normal model

Suppose that $x_1, \dots, x_n \sim N(\theta, \sigma^2)$ with σ^2 fixed/known. Then

$$L(x|\theta) \propto \exp \left\{ -\frac{\sum (x_i - \theta)^2}{2\sigma^2} \right\}.$$

Now with a $\theta \sim N(b, d^2)$ prior, the posterior becomes:

$$\begin{aligned} \pi(\theta|x) &\propto \exp \left\{ -\frac{(\theta - b)^2}{2d^2} \right\} \exp \left\{ -\frac{\sum_{i=1}^n (x_i - \theta)^2}{2\sigma^2} \right\} \\ &= \exp \left\{ -\frac{(\theta - b)^2}{2d^2} - \frac{\sum_{i=1}^n (x_i - \theta)^2}{2\sigma^2} \right\} \\ &= \exp \left\{ -\frac{\theta^2 - 2b\theta + b^2}{2d^2} - \frac{\sum_{i=1}^n x_i^2 - 2n\bar{x}\theta + n\theta^2}{2\sigma^2} \right\} \end{aligned}$$

Conjugate priors

Example: Normal model (contd.)

$$\begin{aligned}\pi(\theta|x) &\propto \exp \left\{ -\frac{\theta^2 - 2b\theta + b^2}{2d^2} - \frac{\sum_{i=1}^n x_i^2 - 2n\bar{x}\theta + n\theta^2}{2\sigma^2} \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left[\theta^2 \left(\frac{1}{d^2} + \frac{n}{\sigma^2} \right) - 2\theta \left(\frac{b}{d^2} + \frac{n\bar{x}}{\sigma^2} \right) \right] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left(\frac{1}{d^2} + \frac{n}{\sigma^2} \right) \left(\theta - \frac{\frac{b}{d^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{d^2} + \frac{n}{\sigma^2}} \right)^2 \right\}\end{aligned}$$

and so

$$\theta|x \sim N \left(\frac{\frac{b}{d^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{d^2} + \frac{n}{\sigma^2}}, \frac{1}{\frac{1}{d^2} + \frac{n}{\sigma^2}} \right).$$

Can express more neatly in terms of precisions $\tau = 1/\sigma^2$ and $c = 1/d^2$:

$$\theta|x \sim N \left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau} \right).$$

Conjugate priors

$$\theta|x \sim N\left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau}\right).$$

Some observations:

1. Note that:

$$E(\theta|x) = \gamma_n b + (1 - \gamma_n)\bar{x}$$

where

$$\gamma_n = \frac{c}{c + n\tau} \quad 1 - \gamma_n = \frac{n\tau}{c + n\tau}.$$

- $\Rightarrow$ **posterior mean is weighted average** of prior mean and $\bar{x}$.
- Weight γ_n is **determined by relative strength of information** in prior and data components.
- If $n\tau$ is large relative to c , then $\gamma_n \approx 0$ and posterior mean $\approx \bar{x}$.

2. **Posterior precision** = **prior precision** + $n \times$ **data precision**

Conjugate priors

$$\theta|x \sim N\left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau}\right).$$

3. The posterior depends on the data only through $\bar{x}$, and not through the x_i . We say that $\bar{x}$ is sufficient for θ .
4. For very large n ,

$$\theta|x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

so the the prior has no effect as the amount of data gets large.

5. As prior standard deviation ($d \rightarrow \infty$) (i.e. $c \rightarrow 0$), then

$$\theta|x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

(more on this when we look at improper priors)

4. and 5. reach the same result as they both have priors that are diffuse with respect to the likelihood (just in two different ways).

Conjugate priors

Some standard prior-likelihood conjugate pairs:

Likelihood	Prior	Posterior
$x \sim \text{Bin}(n, \theta)$	$\text{Beta}(p, q)$	$\text{Beta}(p + x, q + n - x)$
$x_1, \dots, x_n \sim \text{Poisson}(\theta)$	$\text{Gamma}(p, q)$	$\text{Gamma}(p + \sum_{i=1}^n x_i, q + n)$
$x_1, \dots, x_n \sim N(\theta, \tau^{-1})$ (τ known)	$N(b, c^{-1})$	$N(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau})$
$x_1, \dots, x_n \sim \text{Gamma}(k, \theta)$ (k known)	$\text{Gamma}(p, q)$	$\text{Gamma}(p + nk, q + \sum_{i=1}^n x_i)$
$x_1, \dots, x_n \sim \text{Geometric}(\theta)$	$\text{Beta}(p, q)$	$\text{Beta}(p + n, q + \sum_{i=1}^n x_i - n)$
$x \sim \text{NegBin}(r, \theta)$	$\text{Beta}(p, q)$	$\text{Beta}(p + r, q + x)$

You can find more on the [Conjugate Prior](https://en.wikipedia.org/wiki/Conjugate_prior) wikipedia page:

https://en.wikipedia.org/wiki/Conjugate_prior

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Improper priors

Improper prior definition:

A prior distribution, $\pi(\theta)$, such that $\int \pi(\theta) d\theta = \infty$
(likewise, $\sum_i \pi(\theta_i) \rightarrow \infty$ for discrete r.v.)

Example: Normal model with infinite variance.

Recall for normal model: $x_i \sim N(\theta, \sigma^2)$, with $\theta \sim N(b, d^2)$, then writing likelihood and prior precisions as $\tau = 1/\sigma^2$ and $c = 1/d^2$, we have:

$$\theta|x \sim N\left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau}\right).$$

Improper priors

5. As prior standard deviation ($d \rightarrow \infty$) (i.e. $c \rightarrow 0$), then

$$\theta|x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

- ▶ Strength of prior belief is determined by c .
- ▶ Small c means weak prior information (i.e. very large variance).
- ▶ $c \rightarrow 0$ gives the perfectly valid posterior

$$\theta|x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

- ▶ But $c \rightarrow 0$ gives a $\theta \sim N(b, \infty)$ prior distribution, which has density $\pi(\theta) \propto 1$. **Completely uninformative** (i.e. uniform on $\mathbb{R}$).
- ▶ $N(b, \infty)$ is an **improper prior** as $\int_{\mathbb{R}} \pi(\theta) d\theta = \infty$.
- ▶ **OK to use**, as long as $\pi(\theta|x)$ is proper.
(Though see Bayes Factors $\sim$ Lecture Week 8.)
- ▶ If $\int \pi(\theta|x) d\theta = \infty$ then **cannot use this type of prior**.

Bayesian with uninformative vs Frequentist Inference

- ▶ “Uninformative” prior implies no information at all is added, and we are back to frequentist inference. Is this true?
 - **No!** Remember Bayesian inference gives us a distribution on θ , whereas (generally) frequentist inference gives us point estimators (e.g. MLE).
- ▶ Connection between MLE and MAP (maximum a posteriori) estimate:

$$\theta_{MLE} = \arg \max_{\theta} \log L(x|\theta)$$

$$\begin{aligned}\theta_{MAP} &= \arg \max_{\theta} \log \pi(\theta|x) \\ &= \arg \max_{\theta} \log(L(x|\theta)\pi(\theta)) \\ &= \arg \max_{\theta} (\log L(x|\theta) + \log \pi(\theta)).\end{aligned}$$

If we adopt $\theta \sim U(0, 1)$ as our uninformative prior, then we get $\log \pi(\theta) = 0$, i.e. **MLE = MAP**.

- ▶ However, Bayesian inference gives us many other point estimators (e.g. posterior mean), which may not coincide with the MLE (see Tutorials).

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Problems with improper priors

Previous: representing absolute uninformative-ness led to improper priors. However, **this can produce problems.**

Suppose we specify $\theta \sim \pi(\theta) \propto 1$ on $\mathbb{R}$.

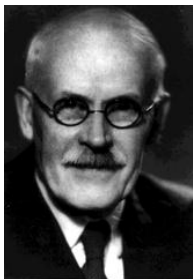
Now **consider the parameter $\phi = \theta^2$** . Using a change of variable on the prior this gives us

$$\begin{aligned}\pi(\phi) &= \pi(\theta) \times \left| \frac{d\theta}{d\phi} \right| \\ &\propto 1/\sqrt{\phi}.\end{aligned}$$

But: if we are ignorant about θ , then **we are equally ignorant about ϕ** .
So we might equally have stated $\pi(\phi) \propto 1$.

Thus prior ignorance (as represented as uniformity of belief) does not translate across scales.

Jeffreys' Prior



Harold Jeffreys
(1891–1989)

Jeffreys' perspective: specification of prior ignorance
should be consistent across 1-1 transformations

Jeffreys' prior defined as:

$$\pi_J(\theta) \propto |I(\theta)|^{1/2}$$

where $I(\theta) = -\mathbb{E} \left[\frac{d^2 \log L(x|\theta)}{d\theta^2} \right]$ is the Fisher information matrix.

Verification (**univariate**):

Suppose $\phi = g(\theta)$ is a monotonic (therefore one to one)
transformation of θ . Then

$$\begin{aligned}\pi_J(\phi) &= \pi_J(\theta) \times \left| \frac{d\theta}{d\phi} \right| \\ &\propto |I(\theta)|^{1/2} \times \left| \frac{d\theta}{d\phi} \right| \\ &= \left| I(\theta) \left(\frac{d\theta}{d\phi} \right)^2 \right|^{1/2} \quad (\det(AB) = \det(A)\det(B))\end{aligned}$$

Jeffreys' Prior

$$\begin{aligned} &= \left| I(\theta) \left(\frac{d\theta}{d\phi} \right)^2 \right|^{1/2} \\ &= \left| \mathbb{E} \left[\left(\frac{d^2 \log L(x|\theta)}{d\theta^2} \right) \right] \left(\frac{d\theta}{d\phi} \right)^2 \right|^{1/2} \\ &= \left| \mathbb{E} \left[\left(\frac{d^2 \log L(x|\theta)}{d\theta^2} \right) \left(\frac{d\theta}{d\phi} \right)^2 \right] \right|^{1/2} \\ &= \left| \mathbb{E} \left[\frac{d^2 \log L(x|g^{-1}(\phi))}{d\phi^2} \right] \right|^{1/2} \\ &= \left| \mathbb{E} \left[\frac{d^2 \log \tilde{L}(x|\phi)}{d\phi^2} \right] \right|^{1/2} = |I(\phi)|^{1/2} \end{aligned}$$

where $\tilde{L}$ is the likelihood under the parameterisation ϕ , as required.

Jeffreys' Prior

Example: Binomial model

Suppose $x|\theta \sim \text{Bin}(n, \theta)$.

Then

$$\log(L(x|\theta)) = x \log(\theta) + (n - x) \log(1 - \theta),$$

so

$$\begin{aligned}\frac{d \log L(x|\theta)}{d\theta} &= \frac{x}{\theta} + \frac{(n-x)}{(1-\theta)} \\ \frac{d^2 \log L(x|\theta)}{d\theta^2} &= -\frac{x}{\theta^2} - \frac{(n-x)}{(1-\theta)^2}\end{aligned}$$

and since $\mathbb{E}(x) = n\theta$,

$$\begin{aligned}I(\theta) = -\mathbb{E}\left\{\frac{d^2 \log L(x|\theta)}{d\theta^2}\right\} &= \frac{n\theta}{\theta^2} + \frac{n-n\theta}{(1-\theta)^2} \\ &= \frac{n}{\theta} + \frac{n}{(1-\theta)} \\ &= n\theta^{-1}(1-\theta)^{-1},\end{aligned}$$

leading to

$$\pi_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2}$$

which in this case is the proper distribution $\text{Beta}(\frac{1}{2}, \frac{1}{2})$.

Jeffreys' Prior

Example: Normal mean model (N_1)

Suppose $X_1, \dots, X_n$ are iid $N(\theta, \sigma^2)$, (σ^2 known).

Then

$$L(x|\theta) \propto \exp \left\{ -\frac{\sum_{i=1}^n (x_i - \theta)^2}{2\sigma^2} \right\} \quad \text{so} \quad \log(L(x|\theta)) = -\frac{\sum_{i=1}^n (x_i - \theta)^2}{2\sigma^2}$$

and

$$\frac{d \log L(x|\theta)}{d\theta} = \frac{\sum_{i=1}^n (x_i - \theta)}{\sigma^2} \quad \frac{d^2 \log L(x|\theta)}{d\theta^2} = -\frac{n}{\sigma^2}$$

$$I(\theta) = -\mathbb{E} \left\{ \frac{d^2 \log L(x|\theta)}{d\theta^2} \right\} = \mathbb{E} \left\{ \frac{n}{\sigma^2} \right\} = \frac{n}{\sigma^2}$$

Hence

$$\pi_J(\theta) \propto 1$$

the improper uniform distribution on $\mathbb{R}$ is Jeffrey's prior.

Jeffreys' Prior

Example: Normal variance model (N_2)

Suppose $X_1, \dots, X_n$ are iid $N(m, \theta)$, (m known), then

$$L(x|\theta) \propto \theta^{-n/2} \exp\{-s/(2\theta)\} \quad \text{and} \quad \log L(x|\theta) \propto -\frac{n}{2} \log \theta - \frac{s}{2\theta}$$

where $s = \sum_i (x_i - m)^2$. Then

$$\frac{\partial \log L(x|\theta)}{\partial \theta} = -\frac{n}{2\theta} + \frac{s}{2\theta^2} \quad \frac{\partial^2 \log L(x|\theta)}{\partial \theta^2} = \frac{n}{2\theta^2} - \frac{s}{\theta^3}.$$

As $\mathbb{E}[(X_i - m)^2] = \text{Var}(X_i) = \theta$, then $E(s|\theta) = \sum_{i=1}^n \mathbb{E}[(X_i - m)^2] = n\theta$.

Hence

$$I(\theta) = -\mathbb{E}\left\{\frac{d^2 \log L(x|\theta)}{d\theta^2}\right\} = -\frac{n}{2\theta^2} + \frac{n}{\theta^2} = \frac{n}{2\theta^2}$$

Jeffreys' prior on $\mathbb{R}^+$ is

$$\pi_J(\theta) \propto \theta^{-1} \quad (\text{improper}).$$

Jeffreys' Prior

Example: Normal variance model (N_3)

Suppose $X_1, \dots, X_n$ are iid $N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$. Then, noting that

$$\begin{aligned}\sum_{i=1}^n (x_i - \mu)^2 &= \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 - 2 \sum_{i=1}^n (x_i - \bar{x})(\bar{x} - \mu) + \sum_{i=1}^n (\bar{x} - \mu)^2 \\ &= (n-1)s_n^2 + 0 + n(\bar{x} - \mu)^2.\end{aligned}$$

where $s_n^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is the sample variance. Then:

$$L(x|\theta) \propto (\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \{ (n-1)s_n^2 + n(\bar{x} - \mu)^2 \} \right]$$

$$\ell = \log L(x|\theta) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \{ (n-1)s_n^2 + n(\bar{x} - \mu)^2 \}.$$

We want to calculate

$$I(\theta) = \mathbb{E} \left(\begin{pmatrix} \frac{\partial^2 \ell}{\partial \mu^2} & \frac{\partial^2 \ell}{\partial \mu \partial (\sigma^2)} \\ \frac{\partial^2 \ell}{\partial \mu \partial (\sigma^2)} & \frac{\partial^2 \ell}{\partial (\sigma^2)^2} \end{pmatrix} \right)$$

Jeffreys' Prior

$$\ell = \log L(x|\theta) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \{(n-1)s_n + n(\bar{x} - \mu)^2\}.$$

$$\begin{aligned}\frac{\partial \ell}{\partial \mu} &= \frac{n}{\sigma^2}(\bar{x} - \mu) & \frac{\partial^2 \ell}{\partial \mu^2} &= -\frac{n}{\sigma^2} \\ \frac{\partial^2 \ell}{\partial \mu \partial (\sigma^2)} &= -\frac{n}{(\sigma^2)^2}(\bar{x} - \mu) = -\frac{n(\bar{x} - \mu)}{\sigma^4} \\ \frac{\partial \ell}{\partial (\sigma^2)} &= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \{(n-1)s_n^2 + n(\bar{x} - \mu)^2\} \\ \frac{\partial^2 \ell}{\partial (\sigma^2)^2} &= \frac{n}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3} \{(n-1)s_n^2 + n(\bar{x} - \mu)^2\}\end{aligned}$$

So we have:

$$I(\theta) = \mathbb{E} \left(\begin{pmatrix} n/\sigma^2 & n(\bar{x} - \mu)/\sigma^4 \\ n(\bar{x} - \mu)/\sigma^4 & -n/(2\sigma^4) + \{(n-1)s_n^2 + n(\bar{x} - \mu)^2\}/\sigma^6 \end{pmatrix} \right)$$

Jeffreys' Prior

$$I(\theta) = \mathbb{E} \left(\begin{pmatrix} n/\sigma^2 & n(\bar{x} - \mu)/\sigma^4 \\ n(\bar{x} - \mu)/\sigma^4 & -n/(2\sigma^4) + \{(n-1)s_n^2 + n(\bar{x} - \mu)^2\}/\sigma^6 \end{pmatrix} \right)$$

Noting that:

- ▶ $\mathbb{E}(X) = \mu$, so that $\mathbb{E}(\bar{X}) = \mu$
- ▶ $\mathbb{E}(s_n^2) = \sigma^2$
- ▶ $\mathbb{E}((\bar{X} - \mu)^2) = \text{Var}(\bar{X}) = \sigma^2/n$

and so

$$\mathbb{E} \left[\frac{(n-1)s_n^2 + n(\bar{x} - \mu)^2}{\sigma^6} - \frac{n}{(2\sigma^4)} \right] = \frac{(n-1)\sigma^2 + \sigma^2}{\sigma^6} - \frac{n}{2\sigma^4} = \frac{n}{2\sigma^4}.$$

Then

$$I(\theta) = \begin{pmatrix} n/\sigma^2 & 0 \\ 0 & n/(2\sigma^4) \end{pmatrix}.$$

And so Jeffreys prior is $\pi_J(\mu, \sigma^2) \propto [\frac{n}{\sigma^2} \times \frac{n}{2\sigma^4}]^{1/2} \propto \sigma^{-3}$ (improper).

Objections to Jeffreys' Prior

Objection 1:

Jeffreys' prior depends on the form of the data (through $L(x|\theta)$).
A prior distribution should not be influenced by what data are to be collected.

Objection 2:

In certain cases, Jeffreys' prior is inconsistent. E.g.

- ▶ N_1 : $x \sim N(\theta, \sigma^2)$ (σ known), $\pi_J(\theta) \propto 1$
- ▶ N_2 : $x \sim N(m, \theta)$ (m known), $\pi_J(\theta) \propto \theta^{-1}$ ($\pi_J(\theta) \propto \sigma^{-2}$)
- ▶ N_3 : $x \sim N(\mu, \sigma^2)$, $\pi_J(\theta) \propto \sigma^{-3}$

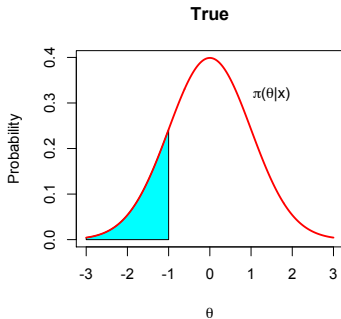
Here $N_3 \neq N_2 \times N_1$.

Jeffreys argues that in this case ignorance about μ and σ^2 should be represented by independent ignorance priors for the two parameters separately.

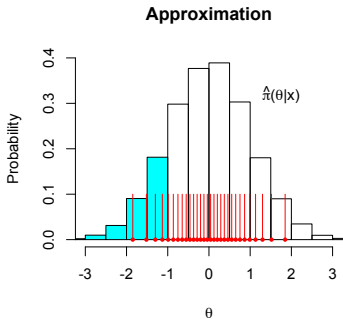
Outline

1. Recap
2. Different types of priors
 - Conjugate priors, improper priors, Jeffrey's priors
3. Monte Carlo Methods
 - Inversion sampling.

Recap: Posterior simulation



$$\begin{aligned}\Pr(\theta < -1) &= \\ \int_{\theta} I(\theta < -1) \pi(\theta|x) d\theta &= \\ \mathbb{E}_{\pi}[I(\theta < -1)]\end{aligned}$$



$$\Pr(\theta < -1) \approx \frac{1}{N} \sum_{i=1}^N I(\theta^{(i)} < -1)$$

where $\theta^{(1)}, \dots, \theta^{(N)} \sim \pi(\theta|x)$.

- ▶ We are interested in ways to draw samples from distributions.
- ▶ $\exists$ many such Monte Carlo algorithms – we will look at several.
- ▶ This week: **Inversion sampling** (the simplest)

Inversion Sampling

Problem: How can we simulate from a univariate distribution $F_X(x)$?

Application of the **probability integral transformation**:

If $X \sim F_X(x)$, then $F_X(X) \sim U(0, 1)$.

Proof: Let U be a random variable given by $U = F_X(X)$.

Clearly $u \in [0, 1]$. The **cumulative distribution function of U** is:

$$\begin{aligned} F_U(u) = Pr(U \leq u) &= Pr(F_X(X) \leq u) \\ &= Pr(X \leq F_X^{-1}(u)) \\ &= F_X(F_X^{-1}(u)) \\ &= u \end{aligned}$$

which is the cdf of a $U(0, 1)$ random variable (when F_X^{-1} exists)

Inversion Sampling

Application of the probability integral transformation (writing $F = F_X$):

If $X \sim F(x)$, then $F(X) \sim U(0, 1)$.

Inversion Sampling

Inverting the probability integral transformation:

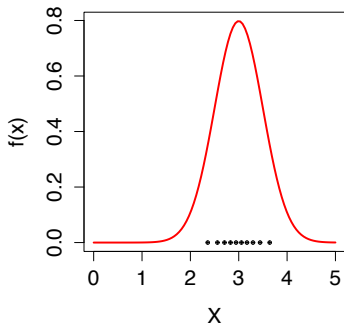
If $U \sim U(0, 1)$ then $F^{-1}(U) = X$.

Hence, to draw n samples from $F(x)$:

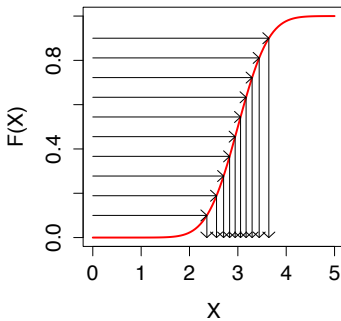
- ▶ Draw $u_1, \dots, u_n \sim U(0, 1)$
- ▶ Set $x_i = F^{-1}(u_i)$.
- ▶ Return $x_1, \dots, x_n$ as *iid* samples from $F(x)$.

Inversion Sampling: Why does it work?

Pdf



Cdf



Let $X = F^{-1}(U)$ where U is uniformly distributed on $[0, 1]$.

The **cumulative distribution function of X** is:

$$\begin{aligned} \Pr(X \leq x) &= \Pr(F^{-1}(U) \leq x) \\ &= \Pr(U \leq F(x)) \\ &= \int_0^{F(x)} 1 du \\ &= F(x). \end{aligned}$$

Example: Inversion Sampling

Problem: Devise a method for simulating a r. v. with the density

$$f(x) = 3x^2 \text{ for } x \in (0, 1).$$

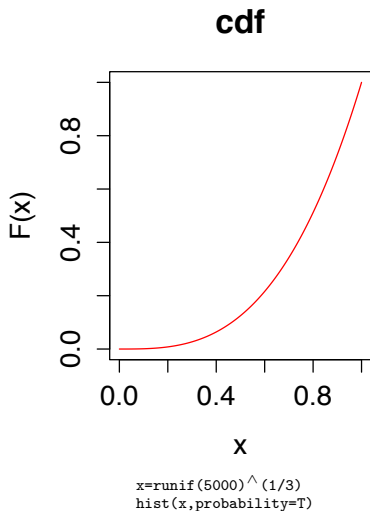
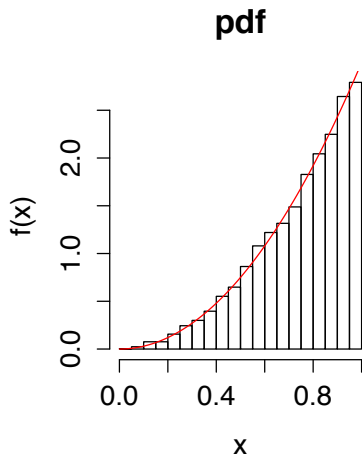
Solution:

First compute the distribution function: $F(x) = x^3$.

Write $F(x) = u$ and then solve for x : $x = u^{1/3}$.

i.e. Generate $u \sim U(0, 1)$ and set $x = u^{1/3} \sim f(x)$.

Example: Inversion Sampling



Example: Inversion Sampling

Problem: Devise a method for simulating a r. v. with the density

$$f(x) = \begin{cases} \frac{(x-2)}{2} & \text{if } 2 \leq x \leq 3 \\ \frac{(2-x/3)}{2} & \text{if } 3 < x \leq 6 \\ 0 & \text{otherwise.} \end{cases}$$

Solution:

First compute the distribution function:

If $x < 2$, $F(x) = 0$.

If $2 \leq x \leq 3$, then

$$\begin{aligned} F(x) &= \int_2^x \left(\frac{z-2}{2} \right) dz \\ &= \left[\frac{(z-2)^2}{4} \right]_2^x = \frac{(x-2)^2}{4}. \end{aligned}$$

Example: Inversion Sampling

If $3 < x \leq 6$, then

$$\begin{aligned} F(x) &= F(3) + \int_3^x \frac{2 - z/3}{2} dz \\ &= \frac{1}{4} + \left[-\frac{3(2 - z/3)^2}{4} \right]_3^x \\ &= \frac{1}{4} + \left[-\frac{3(2 - x/3)^2}{4} + \frac{3}{4} \right] \\ &= 1 - \frac{3}{4}(2 - x/3)^2. \end{aligned}$$

If $x > 6$, then $F(x) = 1$.

Now compute $x = F^{-1}(u)$.

When $x \in (2, 3]$ (equivalently $u \in (F(2), F(3)] = (0, 1/4]$) we have

$$\begin{aligned} (x - 2)^2/4 &= u \\ x - 2 &= \sqrt{4u} \\ x &= 2 + 2\sqrt{u} = F^{-1}(u). \end{aligned}$$

Example: Inversion Sampling

When $x \in (3, 6]$ (equivalently $u \in (F(3), F(6)] = (1/4, 1]$) we have

$$1 - 3/4(2 - x/3)^2 = u$$

$$1 - u = 3/4(2 - x/3)^2$$

$$4/3(1 - u) = (2 - x/3)^2$$

$$\sqrt{4/3(1 - u)} = 2 - x/3$$

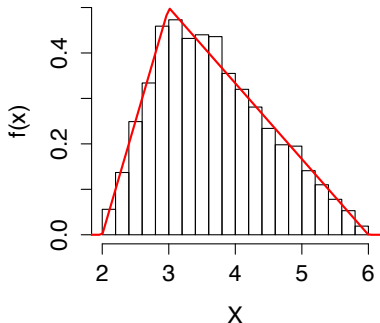
$$x/3 = 2 - \sqrt{4/3(1 - u)}$$

$$x = 3(2 - \sqrt{4/3(1 - u)}) = F^{-1}(u).$$

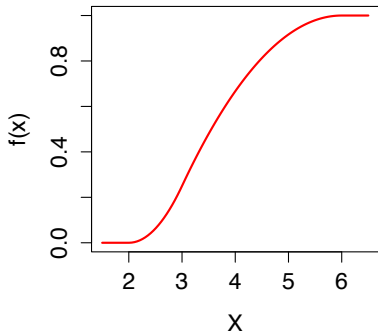
$$F^{-1}(u) = \begin{cases} 2 + \sqrt{4u} & u \in (0, 1/4] \\ 3(2 - \sqrt{4/3(1 - u)}) & u \in (1/4, 1) \end{cases}$$

Example: Inversion Sampling

Pdf



Cdf



```
Finv=function(u) {  
  out=rep(0,length(u))  
  ind=(u<=0.25)  
  out[ind] = 2+2*sqrt(u[ind])  
  out[!ind]= 6*(1-sqrt((1-u[!ind])/3))  
  return(out)  
}
```

```
x=Finv(runif(5000))  
hist(x,probability=T)
```